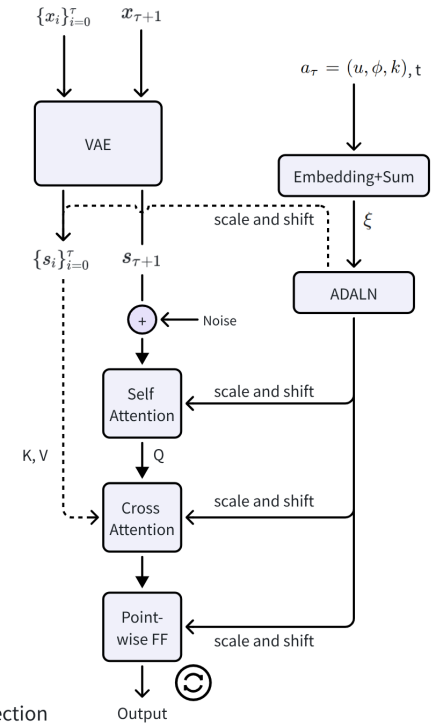
Navigation World Models

> #WorldModel #CV #Navigation #DiT

- Graphical Illustration

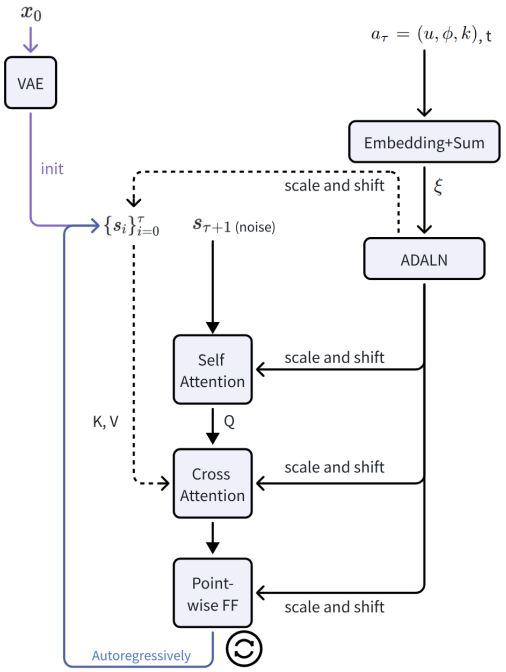
Training Time:



Residual Connection

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{s_{\tau+1}, a_\tau, \mathbf{s}_\tau, \epsilon, t}\left[\|s_{\tau+1} - F_\theta(s_{\tau+1}^{(t)} | \mathbf{s}_\tau, a_\tau, t)\|_2^2\right]$$

Inference Time:



Find a trajectory that minimizes:

$$\mathcal{E}(s_0, a_0, \ldots, a_{T-1}, s_T) = -\mathcal{S}(s_T, s^*) + \\ + \sum_{\tau=0}^{T-1} \mathbb{I}(a_\tau \notin \mathcal{A}_{\text{valid}}) + \sum_{\tau=0}^{T-1} \mathbb{I}(s_\tau \notin \mathcal{S}_{\text{safe}})$$

- Review
- Work Cited
  - Paper
  - Chat History
  - NotebookLM